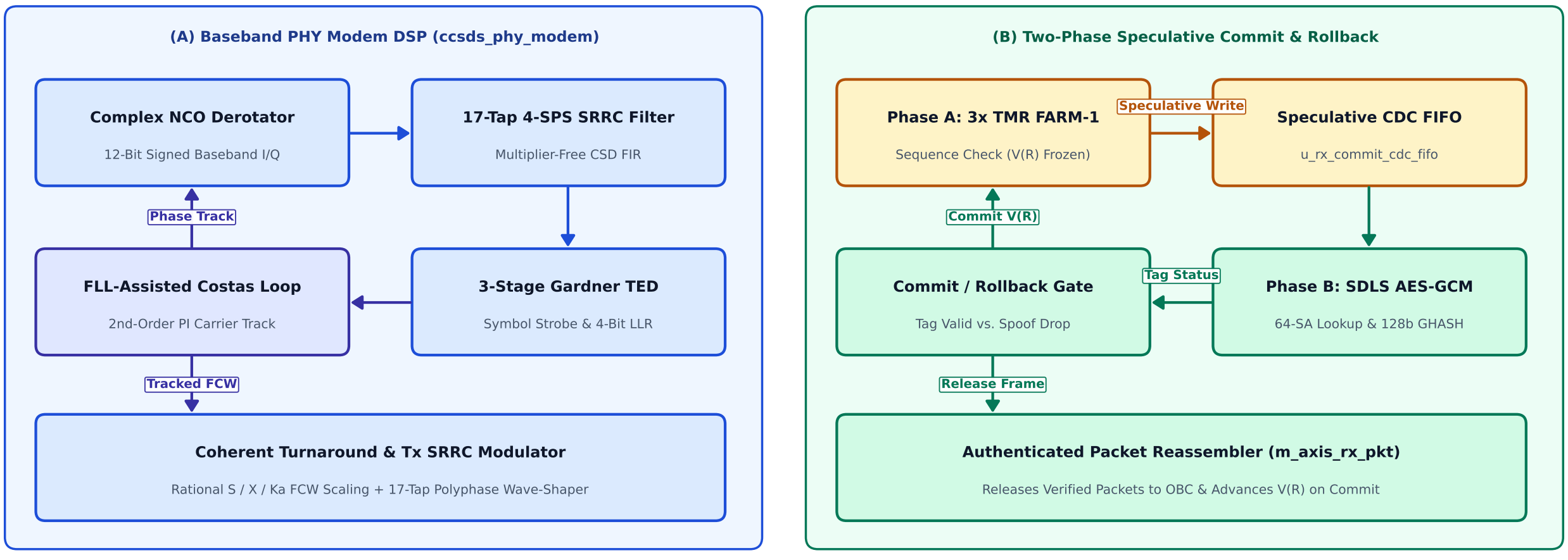
Subsystem Micro-Architectures: (A) Baseband RF Modem DSP & (B) Two-Phase FARM-1 / SDLS Pipeline
(A) Baseband PHY Modem DSP (ccsds_phy_modem)
Complex NCO Derotator
12-Bit Signed Baseband I/Q
17-Tap 4-SPS SRRC Filter
Multiplier-Free CSD FIR
Phase Track
FLL-Assisted Costas Loop
2nd-Order PI Carrier Track
3-Stage Gardner TED
Symbol Strobe & 4-Bit LLR
Tracked FCW
Coherent Turnaround & Tx SRRC Modulator
Rational S / X / Ka FCW Scaling + 17-Tap Polyphase Wave-Shaper
(B) Two-Phase Speculative Commit & Rollback
Phase A: 3x TMR FARM-1
Sequence Check (V(R) Frozen)
Speculative Write
Speculative CDC FIFO
u_rx_commit_cdc_fifo
Commit V(R)
Commit / Rollback Gate
Tag Valid vs. Spoof Drop
Tag Status
Phase B: SDLS AES-GCM
64-SA Lookup & 128b GHASH
Release Frame
Authenticated Packet Reassembler (m_axis_rx_pkt)
Releases Verified Packets to OBC & Advances V(R) on Commit